Supplementary material

lit-acquisition: multi-source, multilingual biomedical literature acquisition with citation-graph traversal

1 Reference set: construction and what it can support

The reference set contains 43 clinically actionable variants and 822 papers. Construction followed a documented automated protocol: for each variant a clinical-style natural-language query was formulated, candidates were pooled from PubMed E-utilities, Europe PMC and Crossref, and a weighted heuristic scored each candidate (gene symbol in title +3; variant identifier +5; disease keywords up to +3; case-report publication type +2; reviews −2; editorials, letters, comments and news excluded). The 30 original variants were annotated in two rounds; 13 further variants were added on 2026-08-14 by `benchmarks/gold_standard/build_variants.py`, which replicates the same protocol without modifying the originals.

Limitations that bound every claim in the paper. (i) The annotations are *heuristic and not expert-verified*, so they contain unquantified false positives and false negatives. (ii) The candidate pool was drawn from channels that *overlap the evaluated system*, a pooling/circularity bias: absolute recall against this set measures agreement with the annotation protocol, not clinical completeness. (iii) Language labels come from provider metadata; 305 of 822 papers have no label, and only German (13) and French (11) carry non-English annotations, so claims about other target languages are not supported by this set. Consequently all reported metrics are *relative* coverage comparisons. `benchmarks/gold_standard/variants.jsonl` is the source of truth; `variants_annotated.jsonl` is a historical V001–V030 file.

2 Configurations and evaluation protocol

Four configurations were compared at a uniform budget of 100 keyword candidates per variant:

- **M1 PubMed-only** — PubMed E-utilities only, English query.
- **M2 multi-source** — all channels, English query.
- **M3 +translation** — adds LLM translation into en/zh/ja/de/fr/ru with language-aware routing.
- **M4n +citation (nested)** — strictly nested ablation built from the frozen M3 ranked list plus citation-graph items recorded in the independent M4 run, deduplicated and appended in traversal order (monotonicity verified; the keyword phase is not re-executed).

Metrics are macro means over the 43 variants: SetRecall (fraction of a variant's reference papers retrieved), Recall@K, Precision@K (fixed denominator K), Precision@Retrieved, and nDCG@K for $K \in \{5, 10, 15, 20, 50\}$. Micro aggregates are also reported in the artifacts. Statistics: two-sided paired Wilcoxon signed-rank tests with zeros dropped, Holm–Bonferroni correction over a pre-declared family of eight tests, matched-pairs rank-biserial effect size, and 10,000 variant-level

percentile bootstrap resamples (seed 42). Live-API run-to-run variance was estimated from two independent executions of the M3 configuration (18/30 identical ranked lists; mean Jaccard 0.891).

Reproducibility of the LLM components. The recorded collection run used `Qwen/Qwen2.5-72B-Instruct` for translation (temperature 0.2, max_tokens 1024, system-prompt SHA-256 prefix `96fe414ab84606dd`) and `BAAI/bge-reranker-v2-m3` as the cross-encoder rerank stage. All of these are recorded in `benchmarks/results/run_manifest.json`. The relevance gate was *disabled* during collection, so the reported metrics are raw retrieval comparisons; the lite edition performs no LLM calls at all.

3 Full metrics

Table 1: Full per-method metrics over 43 variants (macro means). Precision@K uses a fixed denominator K; Precision@Retr. is over the retrieved list.

Method	SetRec.	Rec@5	Rec@20	Rec@50	P@20	nDCG@20	nDCG@50
M1 PubMed-only	0.269	0.076	0.171	0.218	0.155	0.205	0.219
M2 multi-source	0.472	0.044	0.153	0.275	0.147	0.164	0.222
M3 +translation	0.465	0.085	0.276	0.446	0.265	0.301	0.381
M4n +citation (nested)	0.467	0.085	0.276	0.446	0.265	0.301	0.381

4 Significance tests

Table 2: Pre-declared eight-test family: paired Wilcoxon signed-rank with Holm–Bonferroni correction, 10,000 bootstrap resamples (seed 42). r is the matched-pairs rank-biserial correlation.

Contrast	Δ	95% CI	p_{Holm}	r
M2 vs M1 set recall	+0.203	[+0.128, +0.277]	1.08e-04	+0.77
M3 vs M2 set recall	-0.007	[-0.087, +0.069]	1.00e+00	+0.02
M4n vs M3 set recall	+0.002	[+0.000, +0.004]	1.00e+00	+1.00
M4n vs M1 set recall	+0.198	[+0.123, +0.269]	1.93e-04	+0.74
M4n vs M2 set recall	-0.005	[-0.086, +0.071]	1.00e+00	+0.03
M4n vs M1 recall20	+0.105	[+0.035, +0.170]	1.61e-02	+0.51
M4n vs M1 ndcg50	+0.162	[+0.086, +0.234]	4.30e-04	+0.66
M4n vs M3 prec retr	-0.011	[-0.016, -0.006]	1.91e-06	-1.00

5 Comparison with single-channel baselines

Note that the four baselines that only reorder the shared PubMed candidate pool (mean 80.7 candidates: dense, cross-encoder, agentic and multilingual dense) *cannot* exceed M1 on set recall by construction — their SetRecall columns are identical to M1's. The informative comparison is the hybrid BM25+dense+cross-encoder baseline.

Table 3: Single-channel baselines (uniform 20-paper budget over the same PubMed candidate pool unless noted). Significance is versus M4n.

Method	SetRecall	Recall@20	p_{Holm} vs M4n
S1 Dense (bge-m3)	0.269	0.182	3.86e-04
S2 Cross-encoder rerank	0.269	0.171	3.86e-04
S3 Hybrid BM25+dense+rerank	0.421	0.271	1.00e+00
S4 LLM query expansion	0.270	0.173	3.86e-04
S5 Agentic 2-round search	0.269	0.171	3.86e-04
S6 Multilingual dense (bge-m3)	0.269	0.181	3.86e-04
M4n pipeline	**0.467**	**0.276**	—

Table 4: Annotated German/French reference papers recovered per method. Only these two languages carry non-English annotations.

Method	German (of 13)	French (of 11)	Total (of 24)
M1 PubMed-only	0	0	0
M2 multi-source	1	0	1
M3 +translation	11	10	21
M4n +citation (nested)	11	10	21
Best English-only baseline (S3 hybrid)	1	0	1

6 Multilingual recovery

Translation is set-level neutral in aggregate (M3 vs M2: $p_{\text{Holm}} = 1.0$) but changes which papers are reachable, which is the practical argument for language-aware routing.

7 Citation-graph exclusivity

Across all 43 variants, citation traversal contributed exactly two relevant reference papers, one of which appears in no keyword list:

- V002 (BRCA1 c.5266dupC) — *Familial history and prevalence of BRCA1, BRCA2 and TP53 pathogenic variants in HBOC Brazilian patients* (Sci Rep 2022, `10.1038/s41598-022-23012-3`).
- V035 (HFE p.C282Y) — *HFE p.C282Y Polymorphism and Risk of Metabolic Syndrome Components: Systematic Review and Meta-Analysis* (Medicina 2026, `10.3390/medicina62030589`).

The same step costs list precision (Precision@Retrieved -0.011, $p_{\text{Holm}} = 1.9 \times 10^{-6}$), so we present traversal as a bounded complementary channel rather than a headline gain.

8 Provider failures and operational notes

The J-STAGE parse failures were traced to a third-party client that emitted percent-encoded URLs; the channel was replaced by a pure-Python backend and returns real results from v0.4.1 onward. Semantic Scholar anonymous access is aggressively rate-limited from shared/data-centre IPs, which is why an API key is recommended for batch use.

Table 5: Provider failures recorded during the collection session (window 2026-08-12 12:09:28 to 2026-08-12 13:47:18).

Signal	Count
Semantic Scholar HTTP 429	78
HTTP 429 (all providers)	78
J-STAGE parse failures	120
Citation-graph failures	0
Translation calls logged	60

9 Cost measurements

Measurement	Value	Artifact
Full pipeline (composed), median	28.65 s/variant	`timing_43variants.json`
Full pipeline (composed), p90	44.61 s/variant	`timing_43variants.json`
Download rate (end-to-end)	3.99 PDFs/min	`download_throughput.json`
Candidate→PDF conversion	9.6% (18/187)	`download_throughput.json`
Seconds per PDF	15.06	`download_throughput.json`
Peak RSS	85.7 MB	`download_throughput.json`

The timing figure is a composition of the translation-phase run and the citation phase from the same 43-variant session, not a fresh end-to-end run, so that it stays tied to the frozen benchmark lists. The download measurement covers four variants and is a small-sample estimate; it also showed that the download layer returns nothing without `LIT_UNPAYWALL_EMAIL`, because every DOI→open-access route is then skipped.

10 Reproducibility

```
pip install lit-acquisition # lite edition (no LLM calls)
pip install "lit-acquisition[agent]" # + translation, gate, rerank

python benchmarks/run_benchmark.py --collect --limit 100 --cg-budget 20
python benchmarks/make_paper_data.py # regenerates paper numbers
python benchmarks/measure_download_and_memory.py --variants 4 --limit 8
python benchmarks/compose_timing.py
python paper/appnote/make_figure1.py
python paper/appnote/make_supplementary_tables.py
python paper/appnote/check_numbers.py # verifies the manuscript
```

Environment of the recorded run: Python 3.12.3, Linux. Provider API keys were read from the environment; the Unpaywall contact email is required for the DOI→open-access route. Reference-set and result artifacts live under `benchmarks/`; they are excluded from the PyPI distribution and must be taken from the repository.

11 Usage examples

Search across all channels (lite edition).

```
import asyncio
from lit_acquisition import online_acquisition_workflow
```

```
async def main():
    result = await online_acquisition_workflow({
        "query": "MECP2 Rett syndrome case report",
        "action": "search",
        "limit": 20,
    })
    print(result["summary"]) # one-line outcome
    print(result["diagnostics"]) # per-provider status/latency/counts

asyncio.run(main())
```

Multilingual acquisition with downloads (agent edition).

```
from lit_acquisition import multilingual_acquisition_workflow

result = await multilingual_acquisition_workflow({
    "query": "HFE p.C282Y hereditary haemochromatosis",
    "action": "download",
    "language": "multilingual",
    "limit": 30,
    "download_path": "./downloads",
})
```

Citation-graph traversal from a seed DOI.

```
from lit_acquisition import traverse_citation_graph

papers = await traverse_citation_graph(
    seed="10.1038/s41598-022-23012-3", max_depth=1, max_papers=20, direction="both",
)
```